

ALAN NGUYEN PHAM

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Ph.D. Candidate | Soft Robotics | AI-Directed Computational Design | Hands-On Operations

PROFILE

Mechanical engineering Ph.D. candidate with NASA L'SPACE, Sandia, NSF NRT, and WPI soft-robotics experience. Builds ambitious physical systems, teaches 90+ student groups, runs disciplined hardware tests, and keeps complex work organized. Known for fast learning, patient people-facing communication, clean hands-on work, procedure discipline, and calm execution in busy environments.

CORE STRENGTHS

Fast learning | Patient people-facing communication | Clean, organized work habits | Checklist discipline | Reliable execution | Team coordination | Safety awareness | Hands-on troubleshooting | Training/teaching | Documentation/inventory habits

SELECTED EXPERIENCE

Graduate Researcher - Soft Robotics & AI-Directed Computational Design | WPI | Aug 2023-Present

- Lead research on monolithically printed Sarrus/PneuNet cells that transform from planar morphing sheets into cylindrical and continuum robots, crawling/rolling assemblies, and traveling-wave surfaces.
- Directed AI-assisted development of WaveVis by specifying target overhang/wave behavior, connector constraints, UI flow, and output metrics; tested many iterations until it supported CAD and prototype decisions.
- Run pressure, force, displacement, stiffness, and hysteresis tests while maintaining organized prototypes, build notes, data, tools, and parts across fast iteration cycles.

Soft Robotics Graduate Researcher - Project SELENE | NASA L'SPACE | Aug 2025-Present

- Develop modular cable-driven soft robotic arm concepts for terrain navigation and sample collection in remote, hard-to-reach, and extreme exploration settings.
- Lead CAD updates, fabrication planning, 3D-printing decisions, assembly strategy, and structured test planning while guiding undergraduates and coordinating with NASA technical mentors.

Graduate Research Intern - Nonlinear Mechanics and Dynamics | Sandia National Labs | Jun-Aug 2025

- Built analytical, MATLAB multibody, and finite-element submodels for an electromechanical ratcheting mechanism with intermittent contact, friction, springs, shock, and vibration.
- Ran structured parameter sweeps to separate physical response from modeling sensitivity; presented final results supporting more defensible setup choices before full assembly studies.

Teaching Assistant - Stress Analysis, Statics, CAD | WPI | Aug 2023-Present

- Lead weekly sessions and office hours for 90+ student groups; explain difficult engineering material clearly, patiently, and step-by-step without slowing the room down.
- Prepare hands-on activities with professors and teaching teams while keeping sessions organized, respectful, and useful for students with different skill levels.

Hardware Research Assistant - Medical Robotics | WPI Cognitive Medical Technology Lab | Aug-Dec 2023

- Fabricated and assembled low-cost open-source actuation hardware for concentric tube continuum robots; coordinated sourcing, assembly, MATLAB validation, and optical-tracking experiments for two complete systems.

EDUCATION & RECOGNITION

Ph.D. Candidate, Mechanical Engineering - Worcester Polytechnic Institute, expected 2027 | M.S., Mechanical Engineering - California State University, Long Beach, 2022 | B.S., Mechanical Engineering - California State University Maritime Academy, 2019

NSF NRT FORW-RD Fellowship, 2024 | WPI Intercollegiate Soft Robotics Competition, 1st Place, 2024 | \$1,000 NSF NRT professional-development grant

PRACTICAL SKILLS

3D printing; CAD; prototype assembly; hand tools; soldering; equipment setup; testing; troubleshooting; documentation; BOM/inventory tracking; clean shop/lab habits; safety procedures; team training; customer-facing instruction; AI-assisted tool iteration.

Activities: WPI Philharmonic Orchestra (1st violinist); WPI Video Game Orchestra; WPI Tennis Club.

Availability: Flexible part-time, including evenings/weekends upon request.